Junbo (Jacob) Lian

Ph.D. Student, Institute of Operations Research and Analytics (IORA), National University of Singapore
jacoblian@u.northwestern.edu | junbolian.github.io | Google Scholar | ORCID

RESEARCH INTERESTS

Retail pricing; inventory management; large language models for operations research.

EDUCATION

National University of Singapore, Singapore — Jan 2027 – present
Ph.D. in Operations Research and Analytics, Institute of Operations Research and Analytics (IORA)
Advisors: Chung-Piaw Teo and Hanzhang Qin
NUS Research Scholarship

Northwestern University, Evanston, IL, USA — Sep 2025 – Dec 2026
M.S. in Machine Learning and Data Science

Zhejiang A&F University, Hangzhou, China — Sep 2021 – Jun 2025
B.S. in Data Science and Big Data Technology
National Scholarship, Ministry of Education of China (top 0.2% of students nationwide), 2024
Outstanding Undergraduate Thesis, 2025

RESEARCH

Refereed Conference Publications

"ReLoop: Structured Modeling and Behavioral Verification for Reliable LLM-Based Optimization" (with Yujun Sun, Huiling Chen, Chaoyu Zhang, Hanzhang Qin, and Chung-Piaw Teo). *Advances in Neural Information Processing Systems (NeurIPS)*, 2026.
Presented at the POMS International Conference in China, Xi'an, 2026.

Working Papers

"Partial Price Transparency and Demand Learning in Retail Price Competition" (with Hanzhang Qin and Chung-Piaw Teo). SSRN Working Paper No. 7620880.

"Admission Without Answers: Label-Free Certification and Experience Learning for LLM-Based Optimization Modeling" (with Huiling Chen, Hanzhang Qin, and Chung-Piaw Teo). arXiv:2608.15565.

"Reliable Execution and Reusable Certification of Dual-Sourcing Inventory Policies" (sole author). SSRN Working Paper No. 7446638.

Work in Progress

"Repricing a Large Assortment Without Demand Curves" (with Hanzhang Qin and Chung-Piaw Teo). Manuscript in preparation.
Store-level field experiment at Wenzhou Buyi Pharmacy Chain scheduled to begin in March 2027.

Selected Other Publications (machine learning and computational intelligence)

Lian, J., H. Chen, K. Ouyang, Y. Zhang, R. Zhong, H. Chen. 2026. Twisted convolutional networks (TCNs): Enhancing feature interactions for non-spatial data classification. *Neural Networks*: 108451.

Lian, J., T. Zhu, L. Ma, X. Wu, A. A. Heidari, Y. Chen, H. Chen, G. Hui. 2024. The educational competition optimizer. *International Journal of Systems Science* 55(15): 3185–3222.

Lian, J., G. Hui, L. Ma, T. Zhu, X. Wu, A. A. Heidari, Y. Chen, H. Chen. 2024. Parrot optimizer: Algorithm and applications to medical problems. *Computers in Biology and Medicine* 172: 108064.

Lian, J., G. Hui. 2024. Human evolutionary optimization algorithm. *Expert Systems with Applications* 241: 122638.

Other journal articles in evolutionary computation, deep learning, and food engineering are listed on Google Scholar.

HONORS AND AWARDS

Gold Medal, 19th "Challenge Cup" Science and Technology Works Competition, Zhejiang Province 2025
First Prize, National Big Data Challenge Competition of College Students 2022
Two ESI Hot Papers (top 0.1%) and four ESI Highly Cited Papers (top 1%), Clarivate 2024–2026

CONFERENCE PRESENTATIONS

Conference on Neural Information Processing Systems (NeurIPS), poster 2026
POMS International Conference in China, Xi'an 2026

INDUSTRY EXPERIENCE

Wenzhou Buyi Pharmacy Chain Co., Ltd., Wenzhou, China Jan 2025 – present
Chief Analytics Officer

- Lead analytics at a national top-100 pharmacy retailer with about 160 directly operated stores and online channels.
- Built the company's pricing system, including a classification of about 28,000 SKUs and repricing rules for online and offline channels.
- Designed an assortment rationalization tool covering 5,719 drug SKUs (five-level category taxonomy, rule-based delisting with protection rules); the tool is now run by company staff.
- Prepared the store network evaluation for senior management on store closing and opening decisions, and designed the store-level repricing field experiment planned for 2027.

Hangzhou Bossway Testing Technology Co., Ltd., Hangzhou, China Oct 2022 – Jun 2024
Co-Founder and General Manager

- Commercialized a sensor-based mold detection system for Chinese herbal medicine; covered by three national media outlets.
- **First-named inventor** on four Chinese invention patent filings in sensor-based quality inspection, including granted patent CN116465862B (2025).

PROFESSIONAL SERVICE

Ad hoc referee: *IEEE Transactions on Evolutionary Computation*, *Knowledge-Based Systems*, *Expert Systems with Applications*, *Engineering Applications of Artificial Intelligence*, *Applied Soft Computing*, *Swarm and Evolutionary Computation*, *Neurocomputing*, *ISPRS Journal of Photogrammetry and Remote Sensing*, *Journal of Advanced Research*, *Computers in Biology and Medicine*, *Scientific Reports*, *Journal of Big Data*, and 18 other journals.

SKILLS

Computing: Python, SQL, R, MATLAB, C/C++, Gurobi, LaTeX. **Languages:** Chinese (native), English (fluent).